

EXPERIMENT 3 — MONITOR, CAPTURE AND ANALYSE NETWORK PACKETS USING WIRESHARK

Aim

To provide a deeper understanding of network protocol analysis using Wireshark.

Description

Wireshark is an open-source packet analyser used for education, network troubleshooting, communication-protocol development and security analysis. It captures packets travelling through a selected network interface and displays their details for inspection.

Uses

- Troubleshooting network issues.
- Analysing suspicious or malicious network activity.
- Monitoring communication across a network.
- Examining protocol behaviour.
- Supporting digital forensic analysis.

Important Features

- Runs on multiple operating systems.
- Provides a graphical interface for packet analysis.
- Supports packet capture and detailed protocol inspection.
- Can capture traffic from a selected network interface.
- Provides display filters for narrowing the packet list.
- Supports common packet capture formats such as PCAP.

Requirements

Platform	Requirements
Windows	Windows computer, Wireshark, required packet-capture component such as Npcap, network connection
Kali Linux	Kali Linux, Wireshark, Terminal, network connection

PROCEDURE

A. STARTING A CAPTURE

- Step 1 — Open Wireshark.
- Step 2 — Select the required network interface.
- Step 3 — Start packet capture.
- Step 4 — Generate or observe network traffic.
- Step 5 — Stop the capture after sufficient packets have been collected.
- Step 6 — Select packets to inspect their protocol details.

B. WIRESHARK INTERFACE

The Wireshark interface contains a menu/tool area, a packet listing window, a packet-details window and a packet-contents window.

- Packet list: displays captured packets and basic information such as time, source, destination, protocol and length.
- Packet details: displays protocol fields and detailed information for the selected packet.
- Packet contents: displays packet data in hexadecimal and ASCII representations.
- Filter field: allows packets to be filtered according to selected criteria.

C. KALI LINUX INSTALLATION AND LAUNCH

```
sudo apt update
sudo apt install wireshark
wireshark
```

D. DISPLAY FILTERS

Filters can be used to display only packets relevant to the analysis. Examples include:

```
http
tcp
udp
ip.addr == 192.168.1.1
```

E. PACKET ANALYSIS

Select a packet from the packet list to view its protocol details. Source and destination addresses, protocol type, port information and other header fields can be inspected. Filters help reduce a large capture to the traffic needed for the investigation.

F. COMMAND-LINE CAPTURE USING TSHARK

The record notes that command-line capture can be performed with TShark, where available.

```
sudo tshark -i eth0
sudo tshark -i wlan0
```

RESULT ANALYSIS

Wireshark provides detailed visibility into network communication by allowing individual packets to be captured and examined. The captured information can be used to understand protocol behaviour and identify unusual or suspicious traffic.

Captured packets can be inspected using the packet listing and detailed protocol views. Source and destination information can be identified, while protocol-specific packets can be isolated using display filters. Packet contents can also be viewed in hexadecimal and ASCII forms.

From a digital-forensic perspective, packet captures can help investigators understand communication during a network incident and identify relevant activities. The record also notes the usefulness of packet capture for troubleshooting and security analysis.

RESULT

Successfully captured and analysed network packets through a selected network interface using Wireshark. Packet information was inspected using protocol details and display filters.

CONCLUSION

Wireshark is an effective tool for packet capture, network troubleshooting, security analysis, monitoring and forensic investigation. Its capture, filtering and detailed inspection features make it useful for understanding network communication.

OBSERVATIONS

1. Network packets were successfully captured from the selected interface.

2. Packets could be examined using the packet listing and detailed protocol views.
3. Source and destination information could be identified.
4. Different protocols could be observed in the packet list.
5. Protocol-specific packets could be isolated using display filters.
6. Packet details could be inspected at different protocol layers.
7. Packet contents could be viewed in hexadecimal and ASCII forms.
8. The capture can be saved for later analysis.
9. TShark can be used from the Kali terminal where available.
10. Wireshark is useful for network troubleshooting and security analysis.

Practical Flow: Select interface → Start capture → Generate traffic → Stop capture → Select packet → Inspect details → Apply filters → Analyse results.